



Treatment modalities for claw horn lesions and their effects on locomotion scores, gait properties, lesion progression, and nociceptive threshold in dairy cows: A systematic review

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ABSTRACT

This systematic review presents scientific evidence on treatment modalities for claw horn lesions (CHLs) in dairy cows, and their efficacy in restoring normal gait, resolution of claw lesions, and improving nociceptive threshold. A literature search was performed in four databases leading to the synthesis of 10 articles for the final analysis. The types of treatment administered include single (STM), bimodal (BTM) and multimodal treatment modalities (MTM). Locomotion scores (LS) were the most reported outcome measure, followed by the resolution of claw lesion, time to lameness cure, nociceptive threshold and gait properties. Most pairwise comparisons for LS and nociceptive threshold depicted no significant difference between the various treatment groups. Meanwhile, a few pairwise comparisons in terms of resolution of claw lesion and time to lameness cure revealed that cows receiving MTM (claw trimming + hoof block + NSAID) or BTM (claw trimming + hoof block) recorded complete resolution of CHL and significantly less time to recover compared to those treated with an STM (only claw trimming). Despite the various modalities used in CHL treatment, their effects on LS, nociceptive threshold, and gait properties are still unclear. While evidence suggests that MTM and BTM facilitate rapid resolution of CHL and duration of lameness cure, only limited data are available and the association with specific lesions is poorly understood. Several sources of heterogeneity were identified between the studies, particularly duration of follow-up after treatment, LS systems, severity and duration of lameness, and types of treatment administered.

1. Introduction

Lameness in dairy cows remains a global issue that causes impaired animal well-being, a decline in production, and substantial economic losses in the dairy industry (Charfeddine and Perez-Cabal, 2017; Dolecheck and Bewley, 2018). Claw horn lesions (CHLs), particularly sole ulcers, sole haemorrhage, and white line disease, are among the most common non-infectious causes of lameness in dairy cows (Kofler et al., 2024; Roche et al., 2023; Sadiq et al., 2024).

The prognosis and recovery rates of lame cows can be improved by early diagnosis and treatment (Pedersen and Wilson, 2021). Sustainable levels of these on-farm disorders can be achieved by combining 2 approaches: implementing effective farm-specific prevention strategies to reduce the incidence of new cases, and early identification and effective

treatment of clinical cases to reduce the time spent being lame (Vidmar et al., 2021; Mason et al., 2022). Therapeutic intervention is equally pertinent in maintaining the health and welfare of lame cows in dairy settings (Pedersen and Wilson, 2021), as evidenced by the recent increase in the number of research articles on treatment modalities for bovine lameness.

Although a few reviews have presented available evidence on the treatment of lame cows, the extent of recent research focusing on the treatment of CHLs remains underreported. For instance, Laven and Logue (2006) reviewed the understanding of the treatment modalities for digital dermatitis, which is similar to the recent review by Gillespie (2022) on the diagnostic and treatment strategies for these infectious causes of lameness. Hirst et al. (2002) systematically analysed the practical modifiable interventions for lameness in dairy cows while

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other researchers focused on both preventive and treatment measures (Potterton et al., 2012; Huxley, 2013) without considering a systematic approach. Mason et al. (2022) reviewed the use of non-steroidal anti-inflammatory drugs (NSAID) during the treatment of CHLs and the association with locomotion score, lying times, and nociceptive threshold. While the review adopted a systematic approach, NSAID is one of the diverse treatment options for CHLs in dairy cows. The review was also very narrow as duly acknowledged by the authors, stating that the literature on NSAID use for bovine lameness is more extensive relative to their review contents. Moreover, Mason et al. (2022) were principally interested in studies reporting chronic lameness conditions, described in their review as claw horn lameness.

It is well-documented that CHLs induce diverse alterations in dairy cows, such as changes in nociceptive thresholds (Passos et al., 2017), inflammatory biomarkers (Sun et al., 2015), production-related variables (Charfeddine and Perez-Cabal, 2017) and behaviour (Sadiq et al., 2017). These alterations are clinically relevant when investigating lameness response to various treatment options. While it is challenging to include all these outcome measures in a single review, selecting those that constitute the greatest clinical relevance in terms of recovery is of utmost importance. For instance, the review by Mason et al. (2022) considered locomotion scores, lying times and nociceptive threshold as the outcome measures for assessing the response of lame cows to NSAID treatment.

LS is a subjective measure of cows' response to lameness treatment, characterised by low sensitivity for identifying early alterations and hoof pain (Dyer et al., 2007). The assessment of lame cows' response to nociceptive threshold and alterations in lying times are considered more objective outcome measures for identifying recovery from lameness (Mainau et al., 2022). The recovery rate is influenced by a variety of factors such as first or recurring lameness episodes (Shearer et al., 2015), lameness duration (acute vs. chronic) (Thomas et al., 2016), lesion severity (Miguel-Pacheco et al., 2017), and parity (Klawitter et al., 2019). The evaluation of complete healing (horn restoration) of the CHL is also a more objective assessment of lameness treatment relative to LS and gait characteristics.

Despite the available studies on treatment procedures for CHLs in dairy cows, there is still a substantial gap in terms of empirical (scientifically proven) evidence of their efficacy. Although earlier authors concluded that most existing information on the treatment of CHLs was from anecdotal reports gleaned from experts' experiences (Potterton et al., 2012), the number of clinical trials and intervention studies in this research area has increased in the last decade. Hence, this article takes a systematic approach to review the current evidence on the treatment modalities for CHLs in dairy cows, outcome measures used in assessing post-intervention recovery, and factors that may influence the cows' response to the treatment protocols.

This systematic review aims to critically analyse the literature on treatment modalities for CHLs with emphasis on the recovery of lame cows based on locomotion scores, gait characteristics, nociceptive threshold, resolution of claw horn lesion, and time to cure. Therefore, this systematic review investigates if the effectiveness of available treatment modalities for CHLs can be elucidated by assessing locomotion score, gait characteristics, nociceptive threshold, resolution of the lesion, and time taken for lameness cure.

2. Methods

This systematic review was developed according to the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) statement with slight modifications to facilitate data synthesis without meta-analysis (Campbell et al., 2020; Page et al., 2021), also known as SWiM. Registration of the meta-analysis protocol was not attempted.

2.1. Formulation of research questions and definition of terms

This article aims to provide the available evidence on treatment modalities for CHLs in dairy cows, including the parameters used in assessing their efficacy. The PICO instrument was used to synthesise the research questions and literature search process. PICO is a well-established tool for identifying the core components in a review of intervention studies, whereby P = population of interest, I = intervention, C = comparison group, and O = outcomes. In this study, the population of interest was lame dairy cows affected with CHL, and the intervention comprised specific treatments administered. The comparison group entailed lame dairy cows not receiving any treatment, while the outcome measures included locomotion score, gait characteristics, nociceptive threshold, resolution/severity of CHL, and time spent before cure.

Therefore, the research questions developed for this systematic review were as follows:

1. What are the available treatment modalities for CHL in dairy cows?
2. What are the predominant outcome measures used in assessing the efficacy of available treatment modalities for CHL?
3. Do the treatment modalities improve the locomotion score, gait characteristics, nociceptive threshold, resolution of CHL, and time spent before cure?

2.2. Literature search process and eligibility criteria

Relevant research articles were searched from November 23, 2023, to January 23, 2024. The inclusion criteria entailed original research papers, written in English, and published between January 2010 and December 2024. This criterion was based on the surge in studies focusing on the treatment of bovine lameness in the last decade and to address the paucity of previous reviews on the topic. In terms of the study population, articles involving lame dairy cows affected with CHL and intervening with a well-defined treatment modality were considered. Studies published before the year 2010, written in languages other than English, review book chapters, and non-peer-reviewed articles were excluded.

Four search engines namely Web of Science, PubMed, Google Scholar, and Scopus were chosen for the literature search. Aligning with the PICO components, four broad terms namely dairy cows, lameness, claw horn lesions, and treatment were selected in building the search strings as shown in the Supplementary file. The Boolean operators "AND" or "OR" were used in combining the keywords in the search strings, excluding the search string for Google Scholar as only the keywords were used. Definitions of the search terms are provided in Table 1.

2.3. Study selection and screening

The study selection and screening processes were performed by two reviewers (SMB and SSS). The title, keywords, and abstracts of all retrieved articles were independently evaluated by 2 authors (SMB & SZR). Meanwhile, articles that were inaccessible despite sending formal requests to the corresponding authors were not considered for further assessment. Articles fulfilling the inclusion criteria were subjected to full-text screening. The Cohen Kappa was computed to assess intercoder agreement following the preliminary screening (Orwin and Vevea, 2009). Any conflict in the screening process between the authors was resolved by discussion among the authors and the corresponding author was responsible for the final decision.

2.4. Data extraction and manipulation

Two authors participated in the data extraction process. The first author of this manuscript specified the information to be gathered from the articles, while the second author extracted the data accordingly.

Table 1
Definition of search terms employed to search the literature.

Categories	Search terms	Definition
Population	Dairy cows	Cows raised specifically for milk yield
	Dairy cattle	All breeds of cows involved in milk production
	Lactating cows	Cows currently lactating, irrespective of breed
	Periparturient cows	Any breed of cow during the calving period
Intervention	Lame dairy cows	Dairy cows with locomotion disturbance arising from claw horn lesions
	Therapeutic claw trimming	Removal of necrosed or undermined horn and correcting weight bearing on the medial and lateral claws
	Hoof block application	Application of either a wooden or plastic block on the healthy claw to relieve the pressure on the affected claw
	Administration of NSAID	Supportive therapy to reduce inflammation and pain in the affected animal either administering flunixin meglumine, ketoprofen, piroxicam or any NSAID via intravenous, intramuscular routes as recommended by the manufacturers
	Bandaging	Bandaging the affected claw with or without wrapping
Outcomes	Topical medication	Application of topical therapeutic agents such as NSAIDs or only for analgesic purposes.
	Recovery rate	The number of cows achieving non-lame locomotion scores relative to the number of cows treated using a specific treatment protocol
	Nociceptive threshold/ Pain sensitivity	The force required to elicit hoof pain reaction, which is usually measured mechanically using a pressure algometer
	Resolution of claw lesion	Complete resolution and healing of claw horn lesion
	Locomotion score	A subjective scoring method for grading the severity of lameness either before or after treatment
	Lesion size	An assessment method for grading the severity of CHL based on the diameter (cm) of the lesion
	Time for lameness cure	The time taken (i.e., usually measured in days) for the lame dairy cow to achieve a non-lame locomotion score after treatment
	Gait characteristics	Characteristics such as duration of the gait cycle, stance phase, foot load and toe-off. These variables are often assessed using automated devices

First, each article was initially examined for its relevance in answering the research questions, type of research design, well-defined treatment protocol for CHL, and treatment outcomes. We recorded pertinent information about the study such as authors' name and year of publication, study population, treatment protocol used, and main findings (Table 2). Lameness episodes were categorised either as chronic, acute, or both based on the types of CHL and personnel responsible (veterinarian or farmer) for detecting lame cows in the studies. In cases, where such information was lacking, it was described as unknown (See Table 2).

For all the investigated outcome measures, attempts were made to synthesise the results as lame or non-lame as described by Mason et al. (2022). Only the available data in the text were used to recategorize the proportion of animals into lame and non-lame groups. Contrast analyses were performed by comparing all the treatment and control groups in each study.

For studies involving intervention and control groups, pairwise comparisons were conducted based on the assessment of LS < 14 d from the administration of treatment or ≥ 14 d from treatment; the strength

(OR; 95 % CI) and direction of the association was computed for each comparison. In some cases, contrast analyses were not computable because locomotion scores or the outcome measure were presented as normally distributed continuous data. Such data were further analysed using the mean, standard deviation and sample size.

For a study without a defined comparison group, the effect size was computed based on the difference in LS before and after treatment. All the extracted data were recorded in a Microsoft Excel spreadsheet.

2.5. Quality appraisal

We used the *Cochrane Handbook for Systematic Reviews of Interventions* (Higgins and Green, 2011) to assess the quality and risk of bias in the selected articles. Five domains were considered for explicit reporting comprising the sequence generation, allocation concealment, blinding and completeness of data. For the first domain, the use of a randomised sampling method was assessed either by a random number table, tossing a coin, or computerised generated numbers. If such information was lacking, the criterion was recorded as unclear. In terms of allocation concealment, the method used was recorded as adequate if central allocation was conducted, and inadequate if open random allocation was employed.

Regarding the blinding performed in the experimental studies, articles were considered adequate if both the intervention providers and outcome assessors were blinded (i.e., minimal risk of compromised blinding) and inadequate if no blinding or partial blinding was performed. For outcome measures, trials were recorded as adequate if there were no missing data or if the reasons given for missing data were confirmed by the authors of this review to be unlikely associated with the true outcome.

We also assessed if the trial was free from selective reporting whereby studies reporting all prespecified primary measures were judged as adequate and vice versa for those lacking such information. Studies without sufficient information regarding the outcome reporting process were considered unclear. For all the aforementioned criteria, those with insufficient information about each criterion were considered unclear. Demographic characteristics between the treatment groups were also checked for potential bias by considering the cows' baseline information. Two authors (MBS and SSS) performed the quality appraisal independently using a simple form and the results were presented for each study as low, unclear or high, reflecting a low risk of bias, uncertain risk of bias, and a high risk of bias, respectively. The assessment results were compared and any discrepancies were resolved by consulting a third reviewer (WMS).

3. Results

3.1. Search results, extraction of information and study characteristics

The initial literature search yielded a total of 5368 articles. Following the removal of duplicates and screening of the titles and abstracts, 19 articles were included in the full-text reading. Five studies were not included due to one or more of the following reasons; lameness was induced artificially and the type of lameness was arthritis-synovitis (Kleinhenz et al., 2019), lame cows were not enrolled based on CHL (Chapinal et al., 2010), involved only preventive claw trimming to reduce the risk of CHL (Thomsen et al., 2019), and none of the subjective or objective outcome measures were assessed (Miguel-Pacheco et al., 2016; Müller et al., 2023). Three articles were excluded because both CHL and infectious claw lesions were investigated and the results were not reported separately (Groenevelt et al., 2014; Schulz et al., 2016; Chapel et al., 2020), and one study in which secondary data were re-analysed (Miguel-Pacheco et al., 2017). Hence, 10 studies were found eligible and included in the systematic review (Fig. 1).

In line with the inclusion criteria, all the studies reported at least one treatment modality for CHLs. However, there were variations in terms of

Table 2

Basic data of the 10 studies included in the systematic review.

Timeframe	Authors	Study Design	Country	Dairy cows	Lameness acute/chronic	Locomotion score (lameness)	Lesions	Systemic medication	Local treatment	Outcome measure for recovery ("cure")	Control	Time of last evaluation	Limitations
> 14 d	Klawitter et al., 2019	Positive controlled RCT	Germany	64	Acute	≥ 3.0	SU	Ketoprofen and antibiotics	Trimming, block, bandaging	Locomotion score < 3, Lesion size,	Yes	Week 4 (day 28) after treatment	Not blinded
< 14 d	Janßen et al., 2016	RCT	Germany	33	Acute	≥ 3.0	SU, WLD	Ketoprofen	Trimming, block, and bandaging	locomotion scores <3, leg movement	Yes	24 h after treatment	Short term,
> 14 d	Thomas et al., 2016	Positive controlled RCT	UK	164	Chronic	>1/6*	CHL (SUH and WLD)	Ketoprofen	Trimming, block	Locomotion score < 3	Yes	42 ($\pm$ 4) days after treatment	Lesion development was not mentioned
< 14 d	Passos et al., 2017	RCT	Brazil	34	Acute	≥ 4	CHL and infectious claw lesions	None	Trimming	Locomotion score < 3	No	One week after treatment	Short term, lesion development not mentioned, not blinded
> 14 d	Arunvipas et al., 2023	Quasi-experimental design	Thailand	134	Not stated	$\geq 3.0^{***}$	CHL	None	Trimming, block	Locomotion score < 3, absence of pain response upon percussing the lesion, all lesions had been covered by sole growth	Yes	8 weeks after treatment	No randomization was performed, not blinded
< 14 d	Montgomery et al., 2012	One-group pre and post-experimental study	UK	306	Not stated	≥ 2.5	SUH, Bruise, WLD	none	Trimming or block	Locomotion score	No	7 days after treatment	Short-term,
< 14 d	Plüss et al., 2021	RCT	Switzerland	34	Acute	≥ 3.0	CHL	None	Block	Locomotion score < 3, stance phase duration, foot load, toe-off, and mean weight distribution across limbs	Yes	Immediately after treatment	Short term, Lack of treatment for CHL
NA	Mason et al., 2023	Quasi-experimental	New Zealand	241	Acute	$LS \geq 2^{***}$	WLD and SUH	None	Trimming, block	Locomotion score < 3	No	Not specific	Not blinded, no randomization
> 14 d	Thomas et al., 2015	RCT	UK	183 (167)	Acute	>1/6*	SUH, WLD, Other	Ketoprofen	Trimming, block*	Locomotion score ≤ 1	Yes	35 ($\pm$ 4 d) days after treatment	Lesion development was not mentioned
> 14 d	Sadiq et al., 2022	RCT	Malaysia	96 (70)	Acute	>3**	SUH, WLD	Ketoprofen	Trimming, block	Locomotion score < 3**	Yes	28 days	Not blinded

* Detailed description of lameness score; 17/91 blocks lost after 7 days (18, 7 %), loss not documented after 28 days; **** DairyNZ 0–3 Grading System ***[Flower and Weary, 2006](#)** [Sprecher et al., 1997](#), *DairyCo Mobility Scoring System, CHL = Claw horn lesion, WLD = white line disease, SUH = sole ulcer and haemorrhage, RCT = randomised control trial.

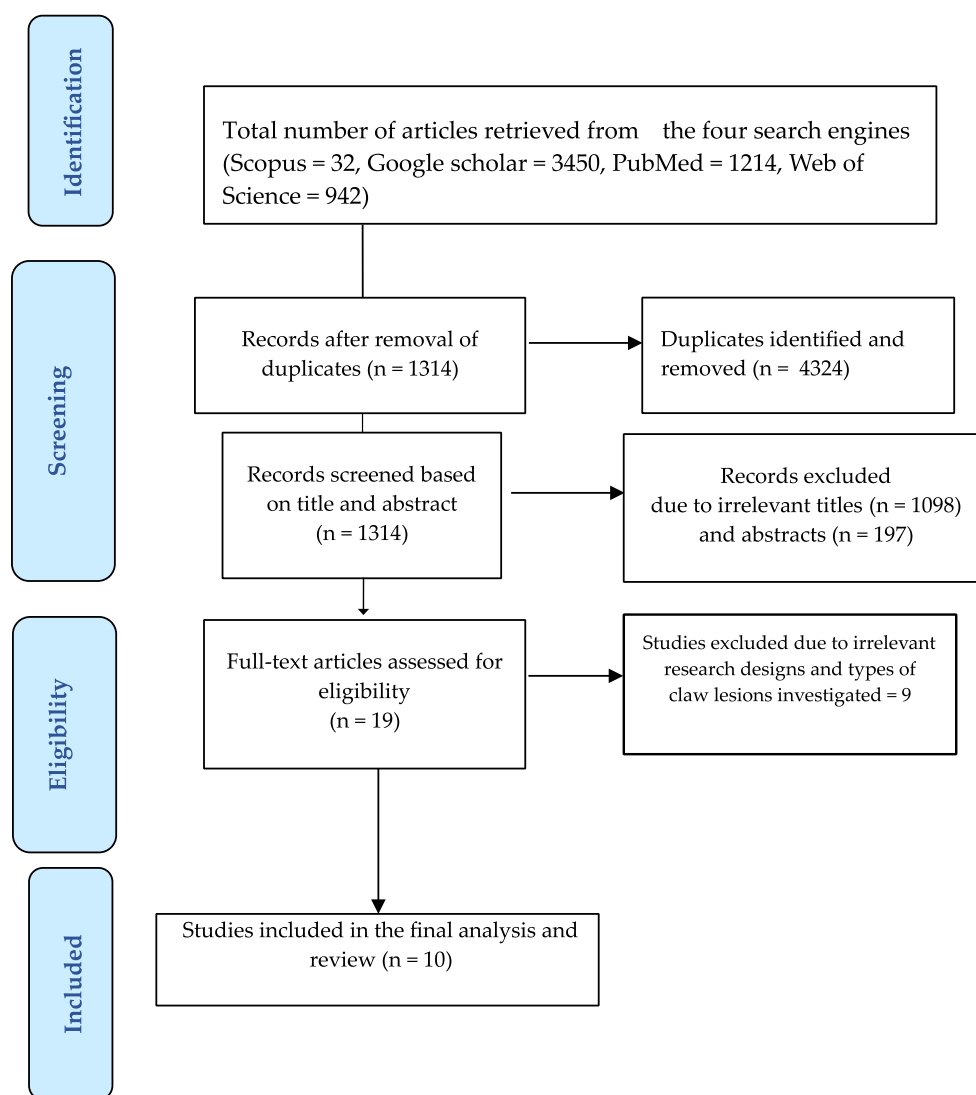


Fig. 1. Literature search process using the PRISMA statement.

study designs, outcome measures, and treatment provided. Detailed information on the objectives, study design, treatment modalities, outcome measures, main findings and clinical relevance of the included studies are presented in Table 2.

Most of the studies were randomised control trials (RCTs) (Thomas et al., 2015, 2016; Janßen et al., 2016; Passos et al., 2017; Klawitter et al., 2019; Plüss et al., 2021; Sadiq et al., 2022; Arunvipas et al., 2023) and three studies were quasi-experimental designs (Montgomery et al., 2012; Passos et al., 2017; Mason et al., 2023). All types of CHLs were investigated in seven of the included studies. Meanwhile, Klawitter et al. (2019) focused on cows affected with sole ulcers and Passos et al. (2017) investigated both CHLs and infectious claw lesions but the results were reported separately (Passos et al., 2017).

Three studies entailed both acute and chronic CHLs (Montgomery et al., 2012; Arunvipas et al., 2023; Mason et al., 2023), two studies involved chronic cases (Klawitter et al., 2019; Plüss et al., 2021), while the remaining five were acute cases (Janßen et al., 2016; Thomas et al., 2015, 2016; Passos et al., 2017; Sadiq et al., 2022). Cows were predominantly housed in freestalls ($n = 9$), and only 1 study enrolled cows that were pasture-based (Mason et al., 2023).

3.2. Outcome measures

Locomotion scores were reported in 8 studies, either as the primary

outcome (Montgomery et al., 2012; Thomas et al., 2015, 2016; Passos et al., 2017; Sadiq et al., 2022; Mason et al., 2023) or used in assessing the time taken for lame cows to recover after treatment (Arunvipas et al., 2023; Mason et al., 2023). Three studies each assessed the resolution or severity of the CHL (Klawitter et al., 2019; Sadiq et al., 2022; Arunvipas et al., 2023) and time to lameness cure (Sadiq et al., 2022; Mason et al., 2023; Arunvipas et al., 2023) as the primary outcomes. Pain sensitivity/nociceptive threshold was assessed in two studies (Passos et al., 2017; Sadiq et al., 2022) and one study for gait characteristics (Plüss et al., 2021).

3.2.1. Post-treatment follow-up

Immediate assessment after intervention was performed by Janßen et al. (2016) and Plüss et al. (2021) since the authors focused on the short-term effects of the treatment on gait characteristics, and physiological and behavioural variables. Meanwhile, most studies performed weekly assessments ranging from one to two weeks (Montgomery et al., 2012; Passos et al., 2017), four weeks (Klawitter et al., 2019; Sadiq et al., 2022; Mason et al., 2023), five to six weeks (Thomas et al., 2015, 2016) and eight weeks after treatment (Arunvipas et al., 2023).

3.2.2. Types of treatment modalities for CHL

One study entailed the use of only claw trimming for the treatment of CHLs (Passos et al., 2017) and only block application on the healthy

claw (Plüss et al., 2021). These interventions were categorised as **single-treatment modality (STM)**. **Bimodal treatment (BTM)**, therapeutic claw trimming and applying a hoof block on the healthy claw, was performed in two studies (Montgomery et al., 2012; Mason et al., 2023). **Multimodal therapy (MTM)** was applied in the remaining 6 articles, which comprised three or more treatment protocols such as claw trimming, administration of NSAID, blocking the healthy claw (Thomas et al., 2015, 2016; Janßen et al., 2016; Sadiq et al., 2022), and bandaging the affected claw (Janßen et al., 2016; Klawitter et al., 2019; Arunvipas et al., 2023). As for the claw trimming method, all the studies used the conventional five-step Dutch technique. Wooden block was utilised in all the studies that considered blocking the healthy claw, except for Thomas et al. (2016) who used plastic blocks. Ketoprofen was the most common NSAID used in pain management when treating cows with CHL.

3.3. Effect of treatment modalities on locomotion scores

The studies considered in this analysis were those that reported locomotion scores as the outcome variable ($n = 7$), of which 6 articles had adequate intervention and control groups and categorical outcomes to facilitate pairwise comparisons. In these studies, 19 contrast analyses covered associations between **MTM** (Thomas et al., 2015, 2016; Janßen et al., 2016; Sadiq et al., 2022; Arunvipas et al., 2023) or **STM** (Plüss et al., 2021) with lame or non-lame status as the outcome (Supplementary Table 2).

The majority of the contrast analyses (18/19; 95 %) reflected no significant difference in the locomotion scores between the groups, as the 95 % CI of the OR for the pairwise comparisons overlapped 1. However, only 1 of the six pairwise comparisons in one study depicted a significant difference between cows administered **MTM** (claw trimming, NSAID, and blocking the healthy claw) and the control group (only claw trimming) (OR = 4.50; 1.09–19.1) (Sadiq et al., 2022).

Contrast analyses could not be performed for 2 studies involving **STM**, because of the different treatment protocols applied (only hoof block vs only claw trimming) (Passos et al., 2017; Plüss et al., 2023) and measurement of locomotion scores as continues rather than categorical data. Notwithstanding, parameter estimates in each study revealed that the 95 % CI of the OR overlapped 1.

The follow-up from treatment to LS measurement varied between studies, with 4 reporting LS outcomes <14 d post-treatment, and 3 reporting LS ≥ 14 days post-treatment. For the latter, 11 of the 14 analyses found that lame cows in the intervention groups (i.e., **MTM**) recorded a higher recovery rate or lower point estimate lameness risk relative to the comparison groups.

3.4. Gait characteristics

Detailed gait analysis was performed only in one study which focused on the effects of hoof block application in cows with CHL (Plüss et al., 2023). The 95 % CI of the OR overlapped 1 in all the contrast analyses, except for the weight distribution comparisons before and after block application in the non-lame group (Supplementary Table 3).

3.5. Nociceptive threshold

Two of the 10 studies reported nociceptive threshold (Passos et al., 2017; Sadiq et al., 2022) and 7 contrast analyses were conducted involving **STM**, **BTM**, and **MTM** (Supplementary Table 4). None of the analyses reflected a significant difference between the groups with the 95 % CI overlapping 0 (Passos et al., 2017) and 1 (Sadiq et al., 2022). For the time frame ≥ 14 days after treatment, 2 of 6 analyses revealed that cows in the treatment group recorded greater nociceptive threshold point estimates relative to those in the control group. The point estimate was also greater after claw trimming compared to the value before the intervention in the study by Passos et al. (2017).

3.6. Resolution of claw lesion

Only 3 articles fulfilling the inclusion criteria reported clinical healing or resolution of CHL before and after treatment (Supplementary Table 5). A total of seven contrast analyses were performed, and only 1 analysis showed a significant difference between the groups (**MTM** with and without bandage), whereby the 95 % CI did not overlap 1. Meanwhile, contrast analyses for other treatments revealed a point estimate with 95 % CI including 0. All the studies entailed ≥ 14 days after treatment, with the lowest point estimate for resolution of CHL in cows that received **MTM** without bandaging compared to the other groups.

3.7. Time taken to recover from lameness

Three studies were eligible for this category (Klawitter et al., 2019; Arunvipas et al., 2023; Mason et al., 2023), but raw data were available for only 2 studies (Supplementary Table 6). The pairwise comparisons entailed **BTM** and **STM** as none of the 2 studies involved **MTM**. Three of the 4 contrast analyses revealed that cows receiving **BTM** (claw trimming + block) spent significantly less time to recover compared to those treated with **STM** (i.e., only claw trimming).

3.8. Quality assessment and reporting guidelines

Based on the five domains in the Cochrane risk-of-bias instrument, the risk of bias ranged from low to high in the reviewed studies (Table 3). A low risk of bias was identified in four studies (Thomas et al., 2015, 2016; Plüss et al., 2023; Janßen et al., 2016). Meanwhile, a moderate risk of bias was identified in 3 studies (Klawitter et al., 2019; Sadiq et al., 2022; Mason et al., 2023), particularly in terms of the randomization process, sample size calculation, and outcome data. Three studies (Arunvipas et al., 2023; Passos et al., 2017; Montgomery et al., 2012) demonstrated a high risk of bias, particularly due to inadequate sequence generation, unclear assessment of cows' baseline characteristics, absence of allocation concealment and blinding process, and unclear measures to address incomplete data.

4. Discussion

This study is among the few attempts to provide a comprehensive analysis of treatment modalities for CHL in dairy cows and their effects on diverse outcome measures. Several authors have highlighted the lack of empirical evidence on the efficacy of treatment protocols for CHL, as most clinicians rely on their field experience when deciding the type of treatment to be administered. This data paucity was reflected in the present review as only 19 articles were eligible for full-text screening out of the several results retrieved for the abstract and title screening. Furthermore, only 10 articles fulfilled the inclusion criteria for the systematic review. The limited evidence and heterogeneity between the studies indicate the dearth of information for a robust analysis of various treatment modalities for CHLs and their efficacy.

Three categories of treatment modalities for CHLs were identified; **STM**, **BTM** and **MTM**. Therapeutic claw trimming was performed in all the studies except for the research by Plüss et al. (2023) which focused only on the use of claw blocks. The type of lameness cases addressed in the studies ranged from acute to chronic types, reflecting the need for proper management. Moreover, therapeutic claw trimming creates an aerobic microenvironment that greatly reduces the likelihood of further complications associated with abscess formation (Shearer et al., 2013).

Nonetheless, the common scenario in farms is that lame cows are not promptly detected as evident in most cases of CHLs. Thus, therapeutic claw trimming and other modalities such as blocking the unaffected claw, and administration of NSAID, with or without bandaging the affected claw were the predominant types of interventions. Additional protocols are required to address severe and chronic lameness since they often involve the deeper tissues of the corium, rendering only claw

Table 3
Quality assessment for studies included in the systematic review.

Study	Adequate sequence generation	Allocation concealment	Blinding	Addressing Incomplete outcome data	Selective reporting bias	Quality assessment/Risk of bias
Klawitter et al., 2019	Yes	Yes	Absent	Yes	Absent	Moderate
Thomas et al., 2015	Yes	Yes	Absent	Yes	Absent	Low
Thomas et al., 2016	Yes	Yes	Absent	Yes	Absent	Low
Plüss et al., 2021	Yes	Yes	Yes	Yes	Absent	Low
Sadiq et al., 2022	Yes	No	Absent	Yes	Absent	Moderate
Mason et al., 2023	Yes	Yes	Absent	Yes	Absent	Moderate
Montgomery et al., 2012	Yes	Unclear	Absent	Yes	Absent	High
Janßen et al., 2016	Yes	Yes	Absent	Yes	Absent	Low
Arunvipas et al., 2023	Yes	Unclear	Yes	Yes	Absent	High
Passos et al., 2017	No	No	Unclear	Yes	Absent	High

trimming insufficient in ameliorating the injury (Shearer et al., 2015).

Locomotion score was the predominant primary outcome in the included studies, particularly in assessing the recovery rate after treatment. Locomotion scoring offers numerous benefits as a non-invasive and simple approach to evaluating the response of lame cows to various treatments (Linstädt et al., 2024). On the other hand, as acknowledged in the recent review by Mason et al. (2022), the variability in locomotion scoring systems and scales constitutes a major challenge in comparing studies reporting LS as the primary outcome. Four different LS systems were employed in all the 10 reviewed studies, which explains the various outcome measures for LS and statistical analyses applied. These discrepancies are further exacerbated by the diverse farm settings and management practices across countries, making it unfeasible to use a standardised LS system as recommended by the International Committee for Animal Recording (ICAR) guidelines (Egger-Danner et al., 2020).

We found significant differences in LS between interventions involving either BTM or MTM vs comparison groups in only 2 (Thomas et al., 2015; Sadiq et al., 2022) of the 7 articles included for contrast analyses. Although the analysis performed in this systematic review depicted the lower bounds of the 95 % CI overlap an OR of 1 (based on dichotomous outcome; $LS < 3$ and $LS \geq 3$) in the study by Thomas et al. (2015), the effect sizes of both studies were large for MTM involving therapeutic claw trimming + NSAID + block compared to only a STM (i.e., only therapeutic claw trimming).

Nevertheless, the differences in the study designs need to be acknowledged. For instance, both Thomas et al. (2015) and Sadiq et al. (2022) enrolled cows that were lame for less than 2 weeks and the latter involved only primiparous cows with good BCS. While the recruitment of acutely lame cows is important for rapid recovery and controlling for cow-level factors may assist in ascertaining the effects of treatments provided, the findings from other studies are equally crucial since most lameness cases on farms are not usually detected within this time frame (Fabian et al., 2014). For instance, the studies by Klawitter et al. (2019) and Arunvipas et al. (2023) involved chronic CHLs, particularly sole ulcers and white line lesions, which are a reflection of the common scenario of lameness cases in dairy farms. The use of claw blocks was found to be protective while bandaging the claw prolonged the recovery process but robust contrast analyses could not be performed because only two arms were recruited in both studies.

It remains unclear if nociceptor threshold is affected by the type of treatment modality for CHLs. Mason et al. (2022) found no strong evidence to suggest that treatment with NSAID alone increases pain sensitivity over time relative to only claw trimming. In the current review, contrast analysis also depicted no significant effect of treatment type on nociceptive threshold. Passos et al. (2017) only recorded a higher nociceptive threshold (indication of lower pain sensitivity) after claw trimming in lame cows affected with CHLs without any control group. Hence, our review also lacks strong evidence as to the effects of available treatments for CHLs on nociceptive threshold in dairy cows. The use of different assessment methods and few studies included in the

analysis might contribute to this finding.

While several studies have investigated the effects of various treatments on the progression of claw lesions in lame cows, only 3 original articles were identified in the present review that focused on cows affected with CHLs (Klawitter et al., 2019; Sadiq et al., 2022; Arunvipas et al., 2023). Although the data were insufficient for a comprehensive analysis, the result suggests that MTM (claw trimming, blocking and NSAID) facilitated a higher cure rate compared to bandaging the affected claw (Klawitter et al., 2019). The available evidence can be strengthened by developing a standardised protocol for assessing the severity of CHLs and recruiting animals that reflect the actual scenario of lameness in dairy farms. Such procedures will improve the objectivity and reproducibility of lesion severity assessment.

As for the time taken before normal gait was restored, BTM (claw trimming and blocking the healthy claw) appeared to facilitate rapid recovery relative to STM such as only trimming the affected claw (Arunvipas et al., 2021; Mason et al., 2023). Our analysis depicted that the mean difference in the time taken before restoration of normal gait ranges from 1 to 2 weeks between the treatment types, but whether better results will be obtained upon administering MTM (i.e., claw trimming + NSAID and Block) and the possible influence of the type of CHL lesion and lesion severity are poorly understood. Thus, more research is required to elucidate if specific MTM for CHL correlates with the lesion resolution and subsequent cure rate.

We found only one study that considered gait characteristics as primary outcomes while investigating the effects of treatment for CHLs (Plüss et al., 2023). This scarcity of data can be linked to the advanced tools required for monitoring gait characteristics in dairies. While the applicability of assessing gait properties for lameness management remains debatable in typical farm settings, these variables will assist in elucidating the mechanisms underlying recovery after treatment.

There are no reasons to suggest that relevant articles were not retrieved upon performing the literature search. First, our search items were relatively broad since we focused on available treatments for CHLs, rather than NSAID as reported in a previous review (Mason et al., 2022). Second, to gather more relevant data, four databases were used for the literature search, followed by performing a manual search but only 10 research articles fulfilled the inclusion criteria to be included in the final analysis.

The literature on the treatment for CHLs is extensive but we aimed to present comprehensive data to measure the recovery rate and efficacy of such treatments. In addition, specific outcome measures namely locomotion scores, nociceptive threshold, gait properties, resolution of the lesion (i.e., lesion severity) and time spent before cure were considered in this review. Selecting these outcome measures is unlikely to affect the search results given that subjective locomotion score is commonly assessed in lameness-related studies.

Ketoprofen was used in all 7 articles whereby NSAID was administered in combination with other treatment modalities. All the studies used the same regimen, 3 day-course of ketoprofen at 3 mg/kg body weight. Therefore, the external validity of our findings regarding the

efficacy of NSAID is restricted to the use of ketoprofen given the lack of data to compare the outcomes of different regimens. A recent review reported an increasing usage of NSAID in production animals worldwide (Laven, 2020), which is also evidenced in the current systematic review. While farmers and animal welfare-conscious advocates may posit that administering NSAID is beneficial on welfare grounds, no substantial evidence could be gleaned from our review that administering ketoprofen, either as a **BTM** or **MTM**, improves gait properties, pain sensitivity, and locomotion scores over time. Nevertheless, using NSAID is strongly recommended to ameliorate the ongoing inflammation and the painful sensation during claw trimming, either in acute or chronically lame cows (Stock and Coetzee, 2015). A well-designed RCT and investigation of sensitive measures of pain are required to ascertain if the use of NSAID is beneficial for lame cows with CHLs.

Quality appraisal was conducted in this review using the *Cochrane Handbook for Systematic Reviews of Interventions*, designed specifically for interventions such as RCTs. The studies subjected to quality assessment demonstrated a low risk of bias ($n = 3$) to a high risk of bias ($n = 7$). These two categories of studies differed mainly in terms of following the reporting guidelines. The objectives and inclusion criteria in all the 10 studies were clear and well defined but only 4 articles explained how their sample sizes were computed. Hence, the study power remains questionable in most studies, further raising questions about the generalisability of the findings. In order to reduce these potential sources of bias, we computed the direction of the association in each study. Notwithstanding, robust data are still lacking to ascertain the actual efficacy of available treatment modalities for CHLs, particularly their effects on locomotion scores, gait properties, nociceptive threshold, and lesion severity.

As highlighted in previous reviews (Muir et al., 2017; Mason et al., 2022), issues regarding sample size and study power are common in cattle lameness studies and other veterinary field research. In addition, the high risk of bias stemmed mainly from inadequate randomization process, lack of baseline assessment of animals' characteristics, and absence of allocation concealment and blinding process, thereby affecting the reliability of results. Adhering to standard reporting guidelines will assist in addressing some of these problems.

5. Clinical relevance and conclusion

The findings from this systematic have important clinical implications, particularly regarding the treatment of CHLs, which are among the most common causes of lameness conditions in dairy cattle. Despite the high frequency with which they occur, we were only able to identify 10 articles (i.e., published in the last 13 years) investigating various treatment modalities for CHLs. This low number of articles reflects the need for more empirical studies and evidence-based data to support clinical practice.

Claw trimming and applying a hoof block were consistent treatment procedures in most of the reviewed studies, which are important approaches to avoid damage to the surrounding healthy tissues and adjust the weight bearing on the affected claw. Another important aspect is pain management, which could be achieved by administering NSAIDs such as ketoprofen and flunixin meglumine. Pain management remains a crucial aspect in the treatment of lame cows (Mason et al., 2022), particularly to address the welfare implications and promote rapid recovery. CHLs such as sole ulcer and white line disease usually develop from local inflammation and insult around the distal phalanx (Newsome et al., 2021), leading to perturbed gait and low nociceptive threshold (Passos et al., 2017). These events reflect the need for anti-inflammatory and analgesic agents, but the clinical relevance is not convincingly demonstrated with the current assessment methods for lameness-associated pain and recovery (Sadiq et al., 2024).

While the use of locomotion scoring constitutes a significant limitation in assessing the recovery of lame cows, only a few studies entailed clinical examination of the CHLs until healing, nociceptive threshold,

and detailed gait properties – which are indeed pertinent to ascertain the effectiveness of available treatment modalities. This also underscores the importance of clarifying the outcome measures such as recovery rate and cure rate. The former reflects the restoration of normal gait and the absence of lameness, whereas cure rate refers to the healing of the specific claw lesion.

Although the use of bandages has not been widely investigated, reports from a well-designed RCT suggest that bandaging the affected claw after corrective trimming offers no significant benefit (Klawitter et al., 2019). When applied properly and free from moisture and manure contamination, bandages and wraps may assist in keeping the lesion clean and protecting the surrounding tissues from opportunistic infections. Meanwhile, there is limited evidence-based data to suggest that such procedures will facilitate rapid recovery compared to when they are not used. Likewise, the use of topical antimicrobial and parenteral therapy for CHLs was sparingly reported in the reviewed studies. These treatment modalities are commonly used by hoof trimmers and veterinarians despite the data paucity to support their usage in clinical practice (Shearer et al., 2015; Arunvipas et al., 2023).

This review found no substantial evidence to support the use of **MTM** over other treatment modalities based on the outcomes investigated; locomotion score, gait properties, nociceptive threshold, resolution of claw lesion and time taken to cure. Although **MTM** (claw trimming, block, and NSAID) and **BTM** (claw trimming and block) depicted better results in terms of healing of the claw lesion and time taken to recover, more clinical trials are required to substantiate the findings.

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CRediT authorship contribution statement

Mohammed Babatunde Sadiq: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Siti Zubaidah Ramanoon:** Writing – review & editing, Resources, Investigation, Formal analysis, Data curation, Conceptualization. **Sharifah Salmah Syed-Hussain:** Writing – review & editing, Supervision, Resources, Funding acquisition, Data curation, Conceptualization. **Rozaihan Mansor:** Writing – review & editing, Supervision, Resources, Funding acquisition, Conceptualization. **Wan Mastura Shaik Mossadeq:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Data curation, Conceptualization. **Nurhusien Yimer Degu:** Writing – review & editing, Visualization, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no conflict of interest.

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Appendix A. Supplementary data

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